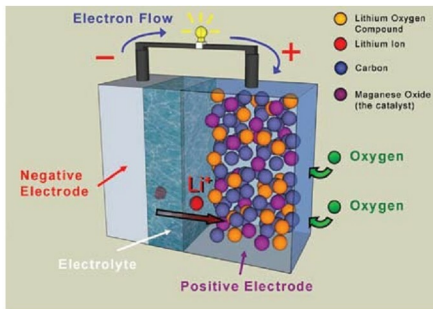


BATTERIES THAT CHARGE ON AIR

Lithium-air batteries use oxygen of the air to charge themselves and are environment-friendly. These also promise tenfold increase in power capacity over lithium-ion batteries



Graphics of a typical Li-air battery

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Lithium-ion batteries power most of the mobile devices like laptops, cellphones and portable PlayStations. Longevity of the battery is important to keep these devices active while on the move. But, unfortunately, lithium-ion batteries are approaching their energy capacity limits.

This has prompted researchers to start finding alternatives. For instance, researchers at the University of St Andrews are developing an air-fuelled battery. In this project, University of Strathclyde and University of Newcastle are partners with University of St Andrews. The research work is funded by the Engineering and Physical Sciences Research Council. This four-year research project, which began in 2007,

is scheduled to end in 2011.

What exists and what's next?

Battery is a diversified field and there exist many types of batteries, each having its own advantages and disadvantages. In the recent past, Li-ion and zinc-air batteries have made a mark in the market.

Zinc-air batteries are electrochemical batteries that charge by drawing oxygen from the air. These have high energy density and are inexpensive to manufacture. They are used in experimental electric vehicles and hearing aids.

Lithium-ion batteries are very successful, but these find applications in limited areas of consumer electronics. Producing about two billion units per

annum, the industry is still growing.

The applications of lithium batteries, however, can be expanded with development of new techniques. To this end, the researches are trying their hands on lithium-air batteries. Lithium-air batteries are showing great results compared to the earlier lithium-ion batteries.

As Peter J. Hall, professor of chemical engineering, University of Strathclyde, explains, "The capacity of Li-air batteries is 10-20 times greater than traditional Li-ion batteries. These will also be considerably cheaper as they contain very little heavy-metal content."

"Since the new batteries use no toxic transitional metal, these are environment-friendly, although there is a definite economic advantage to recycling the lithium content in them." The lithium-air batteries are also going to become lighter, as cathode has been removed from the battery and oxygen from the air is used as the cathode.

What drives Li-air batteries?

Arthur Doble, metal air-battery scientist, SBIR proposal manager, Yardney Technical Products, explains, "Lithium-air batteries breathe oxygen from the air, and allow it to react with the lithium metal anode. These are constructed in layers, like a sandwich, of the lithium metal anode, separator with electrolyte and then the air cathode. The air cathode is the critical part, and that is where oxygen enters to cells and reacts."

Given below is an explanation of Li-air batteries construction from